The Role of Information Framing and Urgency Architecture in App-Based Employment Scams

Approach Paper

Team 22

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1 Research Problem and Questions

App-based employment scams (WhatsApp/Telegram "task-based job" schemes) recruit victims with unsolicited offers, simulate social proof through group chats full of fabricated success stories, and extract upfront payments under manufactured time pressure ("your commission expires in 2 hours," "only 3 slots left today"). These scams are not random in design – they systematically combine two persuasion levers studied separately in neuroeconomics and behavioral security research: **information framing** (how the offer's payoff is worded) and **urgency architecture** (interface/message cues that compress decision time). This project asks:

- **RQ1:** Does gain- vs. loss-framing of a fraudulent job offer affect perceived legitimacy and engagement intent?
- **RQ2:** Does urgency architecture (countdown timers, scarcity language, deadline pressure) independently increase susceptibility, beyond framing alone?
- **RQ3:** Do framing and urgency **interact** – does urgency push targets toward heuristic (System 1) processing in a way that amplifies the framing effect, rather than the two factors acting additively?
- **RQ4:** Can this interaction be traced to a measurable shift in processing style (heuristic vs. deliberative) or decision speed, giving a mechanistic, neuroeconomic account rather than a purely descriptive one?

2 Literature Review

Four threads ground the study, chosen so the review builds toward the gap this project fills rather than simply surveying "scam awareness."

2.1 Prospect theory and framing effects

Kahneman and Tversky's prospect theory (1979) and the classic framing studies (Tversky & Kahneman, 1981) establish that identical outcomes are evaluated asymmetrically depending on whether they are presented as gains or losses relative to a reference point, with losses weighted more heavily than equivalent gains (loss aversion). This asymmetry is the theoretical basis for expecting gain- and loss-framed job offers to produce different compliance rates. A 2025 study in *Behaviour and Information Technology* extends gain-loss framing directly to online scam susceptibility, examining how scam frames, warning frames, and risk perception jointly shape victimization – the closest existing work to this project's framing manipulation, though it does not address employment scams specifically or manipulate urgency alongside framing.

2.2 Persuasion principles and urgency in cyber-fraud

Cialdini's principles of persuasion (authority, scarcity, social proof, liking, reciprocity, consistency) are the standard framework applied to phishing and scam design. Field experiments diverge on which principle dominates: some find authority most effective at lowering suspicion, others find liking and scarcity produce the largest compliance increases, and at least one study finds authority combined with scarcity actually *raises* suspicion rather than lowering it – suggesting these effects are context-dependent and worth pilot-testing rather than assuming. More importantly for this project, large-sample phishing field experiments (Williams, Hinds, & Joinson, 2018; Vishwanath et al.) show urgency cues reliably increase click-through and, counterintuitively, increase message *elaboration* – urgency pulls attentional resources toward processing the message's promised reward rather than toward the cues that would normally raise suspicion. This gives a specific, testable mechanism for why urgency should matter independently of framing.

2.3 Time pressure and dual-process decision-making – the neuroeconomic core

Behavioral and EEG evidence shows time pressure shifts decision-makers toward faster, heuristic, more emotionally weighted processing, altering risk preferences (more risk-seeking for losses, more heuristic reliance for gains) (Zhou et al., 2024; Li et al., 2025). Cognitive-load-induction studies using time pressure and dual-task paradigms provide converging support for dual-system decision theory and offer validated ways to operationalize "cognitive load" as either a manipulation check or a mediator (Deck & Jahedi). Individual differences in coping with time pressure – cognitive ability, cognitive reflection – predict decision quality under pressure, giving a candidate covariate for this study.

2.4 Applied literature on app-based employment scams

Consumer-fraud reports (FTC, EU fraud helpdesks, cybersecurity vendors, 2022–2025) document a consistent script: unsolicited contact via WhatsApp/Telegram, a "training" phase with a visibly rising commission counter, and a mandatory upfront "fee" framed as urgent and time-limited before any payout is released. FTC reporting shows young adults (18–39) and active job-seekers are disproportionately targeted and victimized, which motivates using a student/young-adult sample as ecologically appropriate rather than merely convenient. This layer is largely grey literature rather than peer-reviewed theory, so it is used here for stimulus design and sample justification, not as the theoretical backbone.

Taken together, no existing study crosses gain/loss framing with an urgency manipulation inside an app-based employment-scam script in a controlled factorial design – most framing-and-fraud work targets investment, romance, or generic phishing-email contexts. That gap is this project's contribution.

3 Hypotheses

- **H1 (Framing main effect):** Framing (gain vs. loss) produces a measurable main effect on perceived legitimacy and engagement intent, though the literature does not clearly predict direction in a scam-specific context, making this partly an open empirical question the study is designed to resolve rather than a directional prediction borrowed uncritically from classic framing studies.
- **H2 (Urgency main effect):** High-urgency framing (countdown, limited slots, deadline language) produces higher compliance/engagement intent than low-urgency framing, consistent with phishing field-experiment findings that urgency increases elaboration of the reward message.
- **H3 (Interaction):** The framing effect is larger under high urgency than low urgency, because time pressure shifts processing toward heuristics, a condition under which framing is known to exert more influence.
- **H4 (Mechanism, stretch goal):** The framing $\times$ urgency interaction on compliance is mediated by self-reported heuristic vs. deliberative processing style, or by decision latency as a behavioral proxy for processing depth.

4 Proposed Experimental Design

Given the semester-project time constraint, a between-subjects 2×2 design is not viable. The design is therefore adapted to a **fully within-subjects (repeated-measures) 2 (Framing: gain vs. loss) $\times$ 2 (Urgency: high vs. low)** factorial, in which every participant is exposed to all four conditions. This trades a larger between-subjects sample for repeated observations per person, which is the standard way to preserve statistical power at small N. **Stimuli:** four mock "job offer" chat screens modeled closely on documented real scam scripts (Section 2.4), built around four distinct (but comparable) job types so that repeated exposure does not make the manipulation obvious; core offer content (pay, task description) is held constant across the framing/urgency wording that defines each cell. Presentation order is counterbalanced across participants using a Latin square to control for order and carryover effects. Stimuli are pilot-tested on 5–8 independent volunteers (drawn outside the main N) to confirm each manipulation reads as intended before the main study runs. **Manipulation checks:** after each stimulus, participants rate "How urgent did this offer feel?" and identify whether they read the offer as gain- or loss-framed. **Primary DV:** a behavioral-intent measure – a likelihood-to-register slider, or, if built as an interactive mock page, time-to-decision – collected after every one of the four stimuli. **Secondary DVs:** perceived legitimacy rating, trust rating, and, for H4, either a brief processing-style scale or raw decision latency, also collected per stimulus. **Procedure:** consent $\rightarrow$ four stimulus exposures in counterbalanced order, each followed immediately by its DV and manipulation-check measures $\rightarrow$ funnel debrief probing suspicion $\rightarrow$ full disclosure of the simulated nature of the task.

5 Methodology and Statistical Analysis Plan

Sample size: the project targets $N = 20$**–**30 **participants**. Because the design is fully within-subjects, each participant contributes all four condition observations, yielding **80–120 total data points** rather than 20–30. Repeated-measures designs achieve substantially higher power per participant than between-subjects designs of the same N, because each person serves as their own control and between-person variance is removed from the error term; at $N = 25$ with four repeated observations, a repeated-measures ANOVA is reasonably well powered (power $\approx .80$) to detect medium-to-large within-subject effects ($f \approx 0.30$–0.40), though it remains underpowered for small effects. This is an **exploratory, pilot-scale study** , with results reported as preliminary evidence and effect-size estimates rather than definitive confirmatory tests, and with a recommendation that a larger between-subjects replication follow if effects are promising.

Primary analysis: a two-way **repeated-measures ANOVA** (Framing $\times$ Urgency, both within-subjects) on the primary continuous DV, using Greenhouse–Geisser correction if sphericity is violated. **If the DV is binary** (e.g., would-register yes/no), a **mixed-effects logistic regression** (generalized linear mixed model with a random intercept per participant) is used instead of ordinary logistic regression, since observations from the same participant are not independent; this can be fit with `statsmodels.genmod` (GEE) or `pymer4`/`lme4`-style formula APIs in Python.

Manipulation-check analysis: paired-samples t-tests (or the non-parametric Wilcoxon signed-rank test, given the small N) confirming each factor was perceived as intended across the four within-subject conditions.

Mediation analysis (H4): if pursued, a within-subject mediation approach (e.g., multilevel mediation with participant as a random effect) tests whether processing style or decision latency carries the framing $\times$ urgency effect onto compliance; given the small sample this should be treated as exploratory and interpreted cautiously.

Covariates: age, prior exposure to or victimization by scams, and optionally a brief Cognitive Reflection Test score, collected once per participant and entered as between-subject covariates in the mixed model, acknowledging that with $N \leq 30$ there is limited power to detect covariate effects reliably.

Reporting: all tests accompanied by effect sizes – partial η^2 for the repeated-measures ANOVA, Co-

hen's d_z for paired comparisons, and odds ratios for the mixed logistic model – alongside confidence intervals, since with this sample size point estimates should be interpreted with appropriately wide uncertainty bounds rather than relying on significance thresholds alone.

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